

AI Health Monitoring Guide

Final Project Report

Project Type	AI-powered health monitoring assistant using LightGBM + GenAI
Main Application File	health_monitor_FINAL.py
Requirements File	requirements_health_monitor_FINAL.txt
Core Features	Risk prediction, explanation, weekly trend, voice summary, Q&A, uploaded report summarization

Abstract

AI Health Monitoring Guide is a LightGBM-based health risk monitoring system designed to help users interpret day-to-day health readings more meaningfully. The application combines structured machine learning with GenAI-assisted outputs to move beyond simple threshold alerts. The user can select a patient, enter current readings, compare them with recent history, and obtain a predicted risk label. The system further supports natural-language explanation, weekly trend insight, voice summary, question-answer assistance, and uploaded health-report summarization. The final solution is implemented as a Streamlit application, making it suitable for academic demonstration and for future extension into a broader patient-monitoring assistant.

1. Problem Statement

Many senior citizens and caregivers in senior living communities record health values such as blood glucose, blood pressure, pulse, cholesterol, and comorbidity status, but interpreting these readings correctly is difficult without medical support. In such settings, risk usually depends not on one isolated reading but on short-term trends across several days. A delayed understanding of worsening blood pressure, sugar, or recurring abnormal values can lead to slower intervention and weaker day-to-day monitoring.

This project addresses that gap by building an AI-powered health monitoring guide focused on elderly users, caregivers, and senior living environments. The system not only predicts a current risk label from today’s readings and recent history, but also explains the result in simple language, provides weekly insight, supports voice-based accessibility, and summarizes uploaded medical reports so that health information becomes easier to understand and act upon.

2. Objectives

- Predict a health risk label from current readings and recent historical data.
- Engineer history-aware features using the previous 3–5 days instead of relying only on today’s values.
- Provide clear explanation and weekly insight using GenAI layers.
- Offer accessible interaction through voice summary and a Q&A assistant.
- Extend the system with uploaded image/document summarization for health-related reports.

3. Dataset and Input Variables

The project uses a patient-level longitudinal health dataset with daily readings. The key variables include Patient ID, Date, Day, Age, Gender, BMI, Blood Glucose, Systolic BP, Diastolic BP, Pulse, Cholesterol, Diabetes, Hypertension, and Heart Disease. Where an explicit risk label was not available, rule-based health-risk logic derived from medical-style thresholds was used to generate labels for supervised learning.

Category	Variables	Purpose
Patient profile	Patient ID, Age, Gender, BMI	Identify and contextualize the

		user
Daily vitals	Blood Glucose, Systolic BP, Diastolic BP, Pulse, Cholesterol	Drive the prediction and explanation
Health history	Diabetes, Hypertension, Heart Disease	Capture baseline risk and comorbidities

4. Feature Engineering and Risk Logic

A major strength of the project is history-aware feature engineering. Instead of relying only on the current day, the system calculates previous 3-day and 5-day averages for the major numeric readings. It also measures day-to-day change, difference from recent averages, and count-based flags such as repeated high glucose days or high blood-pressure days. The rule-based risk logic translates vital signs into interpretable bands such as blood-pressure stage, glucose band, cholesterol band, and comorbidity count. These engineered features make the model more robust and help the explanation layer remain grounded in actual health patterns.

5. Model Development

Multiple models were tested earlier in experimentation, but LightGBM was selected as the final model because it performed strongly on structured tabular data and produced highly competitive results. The final model uses engineered health and trend features as inputs and predicts the health-risk label. LightGBM was chosen for its speed, efficiency, and strong performance on the classification task.

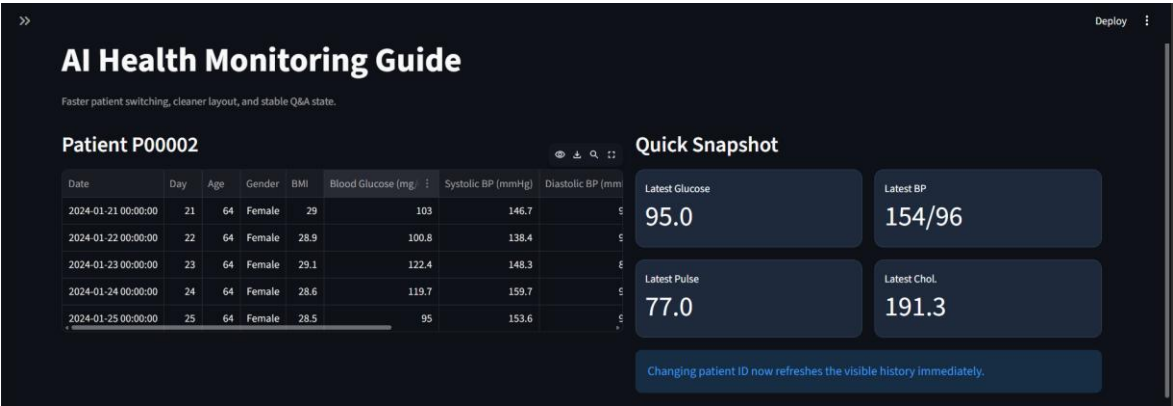
Model Design Choice	Final Decision
Core predictive model	LightGBM classifier
Input design	Current readings + 3-day/5-day history features
Output	Predicted health-risk label
Application format	Streamlit-based interactive interface

6. System Features

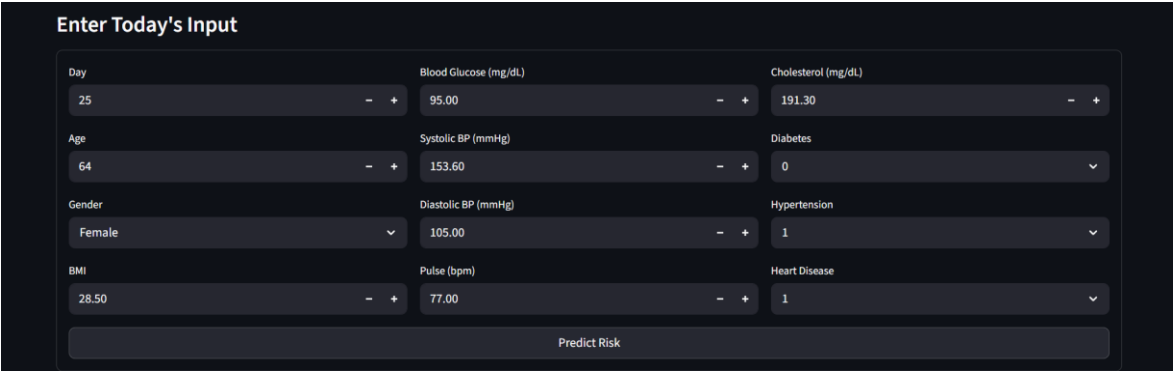
- **Risk Prediction:** Predicts the current health-risk label from today’s input and recent history.
- **Gemini Explanation:** Explains the predicted result in simple language instead of only showing a label.
- **Weekly Trend Insight:** Summarizes what improved, worsened, and what needs attention across recent days.
- **Voice Summary:** Converts the explanation into a voice-friendly summary for easier accessibility.
- **Q&A Assistant:** Allows the user to ask follow-up questions related to risk, trend, diet, and next steps.

- **Health Report Summarizer:** Lets the user upload a health-related image or document and receive a concise summary.

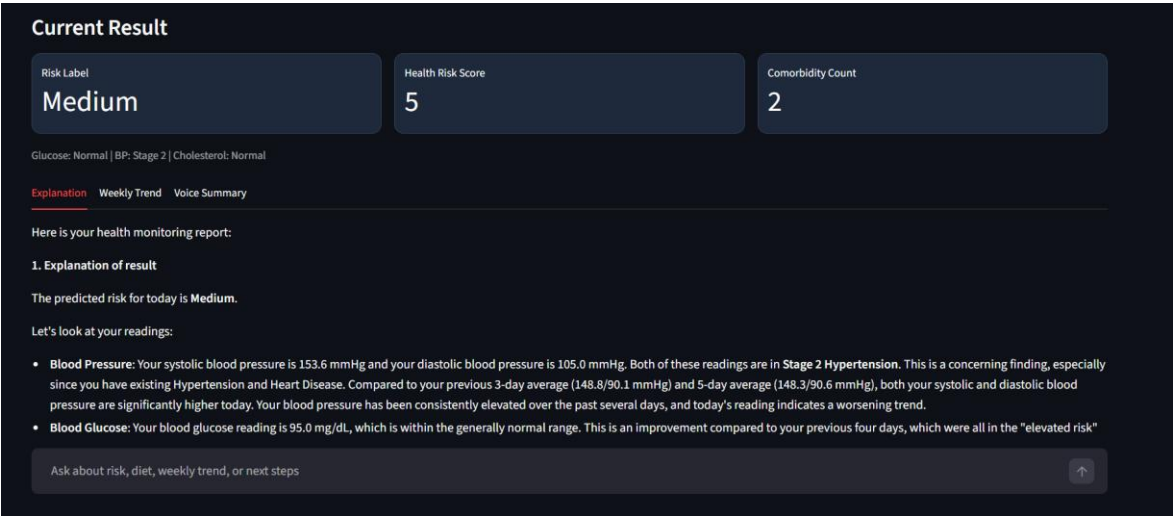
Application interface snapshots



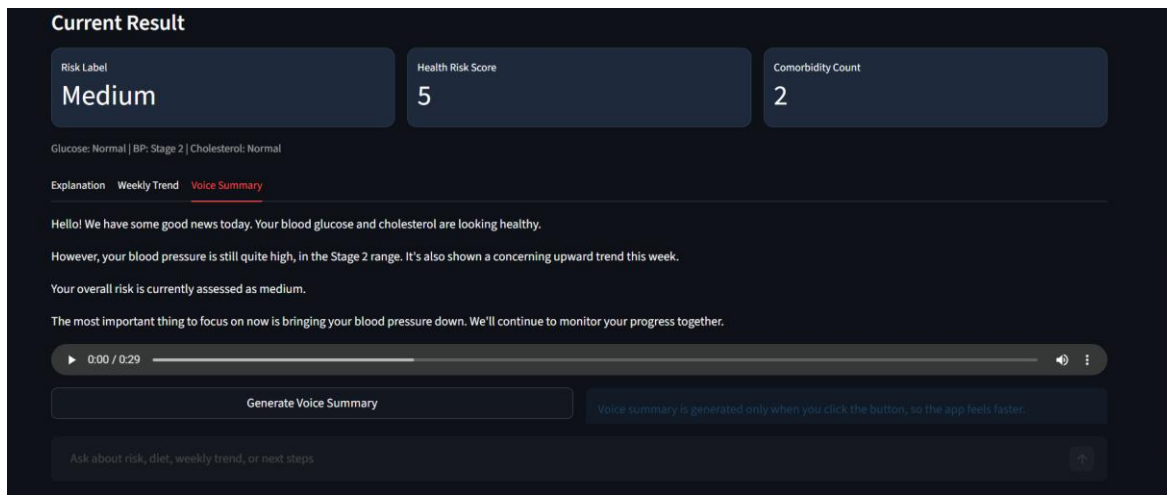
Interface 1: Patient history and quick snapshot dashboard.



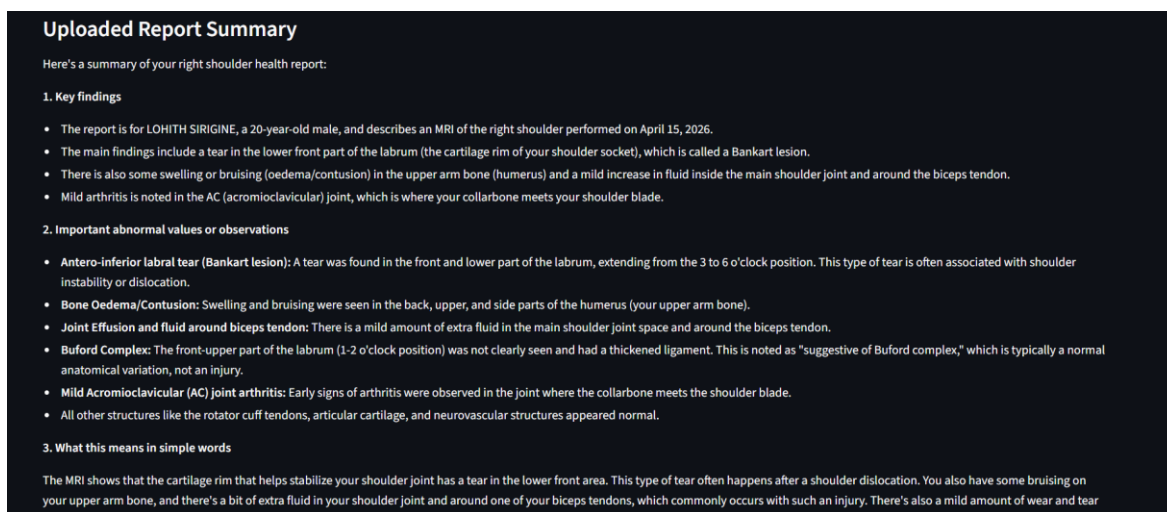
Interface 2: Today's input form for current health readings.



Interface 3: Predicted risk label with explanation and Q&A assistant.



Interface 4: Voice summary interface for accessible output.



Interface 5: Uploaded report summarization for health-related images and documents.

7. Application Workflow

1. The user selects a patient ID from the sidebar.
2. The app displays the past five days of readings and a quick snapshot.
3. The user enters today's health values and clicks Predict Risk.
4. The model predicts the risk label and generates explanation/trend outputs.
5. The user can optionally generate a voice summary and ask follow-up questions.
6. The user can upload a health report image or PDF to receive a summarized interpretation.

8. GenAI Layer Components

- Explanation layer: converts the model output into simple health-oriented language. And also has document summarization layer

- Weekly trend insight: summarizes improvement, deterioration, instability, and attention areas.
- Voice summary: creates a spoken version of the current result for easier listening.
- Q&A layer: allows follow-up questions about the predicted result and patient context.
- RAG layer has also been used for generating better explanations. These layers improve usability, especially for non-technical users.

However, the system should still be described as an AI-powered health monitoring assistant rather than a fully agentic AI system, because it does not autonomously plan and act without user control.

9. Strengths and Limitations

Strengths

- Combines machine learning and GenAI in a clear workflow.
- Uses historical context instead of only single-day values.
- Gives both structured output and natural-language interpretation.
- Includes voice and Q&A features for a better demonstration experience.

Limitations

- The GenAI layer depends on API quota and connectivity.
- This is a support system and not a substitute for medical advice or diagnosis.

Future scope

- Add multilingual voice output and multilingual Q&A.
- Create doctor/caregiver reports automatically.
- Improve offline fallback logic when API services are unavailable.
- Add scheduling, reminders, and more autonomous workflow handling.

10. Conclusion

AI Health Monitoring Guide demonstrates how a structured ML model can be strengthened with practical GenAI layers to create a more complete assistant rather than a plain classifier. By combining risk prediction, explanation, trend summarization, voice output, interactive Q&A, and uploaded report summarization, the project becomes more useful, more understandable, and more demo-ready. The final solution is appropriate for academic presentation and provides a solid base for future extension into a more advanced health-monitoring platform.